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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																												
B									B																																												
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-28 20kV Central Office Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																												
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F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="3"> ABB EPMV PT. ABB Sakti Industri </td><td rowspan="3">COVER SHEET SLD & GA Drawing MVD-28 20kV Central Office</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB EPMV PT. ABB Sakti Industri	COVER SHEET SLD & GA Drawing MVD-28 20kV Central Office	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.		Cont. 16	F
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F					Date Prep.	15.11.2018	PROJECT TITLE							
	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant		ABB	EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-28 20kV Central Office	Scale		= H00
	0	First Issued	19.11.2018	YR	Checked	AW						Customer	PT Multimas Nabati Asahan	& GAD
							PT. ABB Sakti Industri		Drawing Nbr.			Sh.	2	
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES						
	SWITCHGEAR VERSION = UNISEC				COLOUR OF FRONT DOORS = RAL 7035						
	RATED VOLTAGE (Un) = 24 kV				SIDEWALLS PAINTED = YES						
	OPERATING VOLTAGE = 20 kV				SIDES = L+R						
	RATED FREQUENCY (fr) = 50 Hz										
	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up) = 125 kV										
B	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud) = 50 kV										
	RATED CURRENT OF MAIN BUSBARS (Ir) = 630 A										
	RATED SHORT-TIME WITHSTAND CURRENT (Ik) = 20 kA										
	RATED PEAK WITHSTAND CURRENT (Ip) = 50 kA										
	RATED DURATION OF SHORT CIRCUIT (tk) = 1 s										
	INTERNAL ARC CLASSIFICATION (IAC) = AFLR / 16kAx1s										
C	AMBIENT CONDITION = NORMAL										
	AMBIENT AIR TEMPERATURE = -5°C...+40°C										
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED) = IP3X										
	DEGREE OF PROTECTION WITH OPEN DOORS = IP2X										
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua) = 230VAC										
	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua) = 230VAC										
D	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua) = 230VAC										
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS = 450/750V										
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS = PVC										
	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS = 2.5 mm²/BLACK										
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS = 1.5 mm²/BLACK										
	CROSS-SECTION OF CONDUCTORS FOR MOTORS = 2.5 mm²/BLACK										
E	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS) = 1.5 mm²/BLACK										
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS = 1.5 mm²/BLACK										
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE = 2.5 mm²/BLACK										
	COMMUNICATION CABLE = N/A										
F											
1	Issued for Approval	18.12.2018	YR	Date Prep. 15.11.2018 Prepared YR	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	GENERAL CHARACTERISTICS SLD & GA Drawing	Scale Customer	PT Multimas Nabati Asahan	= H00 & GAD
0	First Issued	19.11.2018	YR	Checked AW				MVD-28 20kV Central Office	Drawing Nbr.		Sh. 3
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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION		-CA		CAPACITORS		A														
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION		-CA1		CAPACITOR FOR FRONT VENTILATOR																
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		-CA2		CAPACITOR FOR REAR VENTILATOR																
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION		-EA1		LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.																
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION		-EA2		LIGHTING LAMP LOCATED IN CABLE COMPARTMENT																
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION		-EA3		LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE																
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION		-EA4		LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT																
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION		-EB1		HEATER LOCATED IN CABLE COMPARTMENT																
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION		-EB2		HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT																
	-BGK		POSITION SWITCH OF KEY LOCK		-EB3		HEATER LOCATED IN MOTOR CABINET																
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1		-EB4		HEATER FOR VENTILATION TEST		B														
	-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS		-EB5		HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.																
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION		-EB7		HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE																
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION		-EB8		HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE																
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION		-FA1		SURGE ARRESTER LOCATED ON PHASE L1																
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION		-FA2		SURGE ARRESTER LOCATED ON PHASE L2																
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION		-FA3		SURGE ARRESTER LOCATED ON PHASE L3																
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION		-FCD		FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS																
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION		-FCF1		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1																
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION		-FCF2		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2																
C	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION		-FCF3		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3		C														
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION		-FCM1		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER																
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION		-FCM2...-FCM10		MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS																
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR		-FCM11		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1																
	-BGV2		POSITION SWITCH OF REAR VENTILATOR		-FCM12		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2																
	-BM		HYGROSTAT		-FCM13		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3																
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-FCT		THERMAL OVERLOAD RELEASES																
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GA		GENERATORS																
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ1		FRONT VENTILATOR																
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ2		REAR VENTILATOR																
D	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER		-GQ3		VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK		D														
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2		-KFA1		AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE																
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3		-KFA2		AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE																
	-BR		FLAME DETECTORS, SMOKE DETECTORS		-KFA3....-KFA9		AUXILIARY RELAYS OR CONTACTORS																
	-BT		THERMOSTAT		-KFC		CLOSING RELAYS OR CONTACTORS																
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS		-KFI		INTEGRATED CIRCUITS																
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS		-KFL		LOCKOUT RELAY																
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM		-KFM		MICROPROCESSORS																
	-BU		BUCHHOLZ RELAY		-KFN		ANTIPUMPING RELAYS																
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING		-KFO		OPENING RELAYS OR CONTACTORS																
E	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING		-KFP		PROGRAMMABLE LOGIC CONTROLLERS (PLC)		E														
	-BZ,-BZ4		MULTIFUNCTION SENSORS		-KFR		RECLOSING RELAYS																
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1		-KFS		SYNCHRONIZING DEVICES																
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS																
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3		-KFZ		CONTROLLERS																
	-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER		-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD																
	-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1		-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2																
	-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2		-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS																
	-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS																				
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

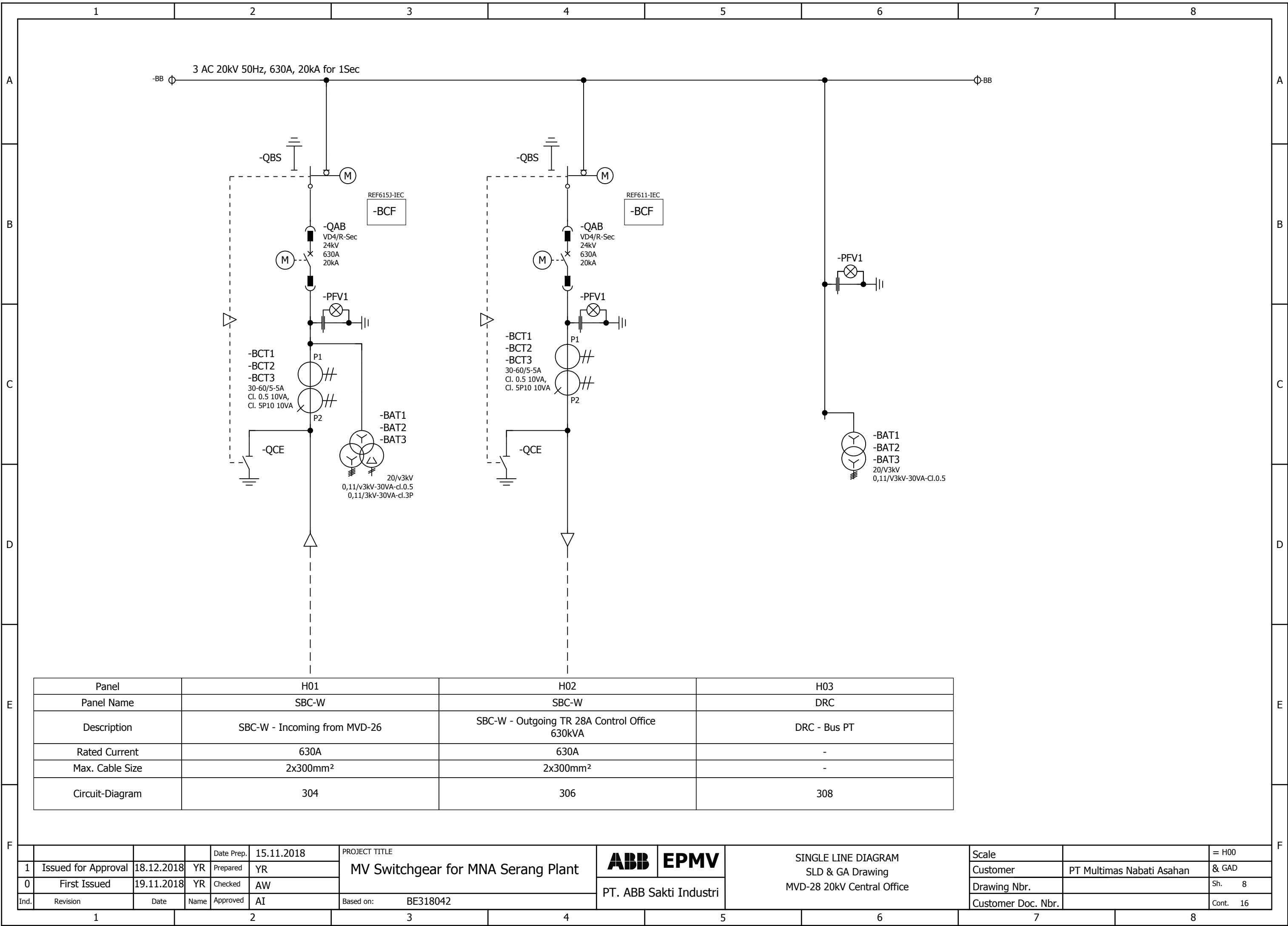
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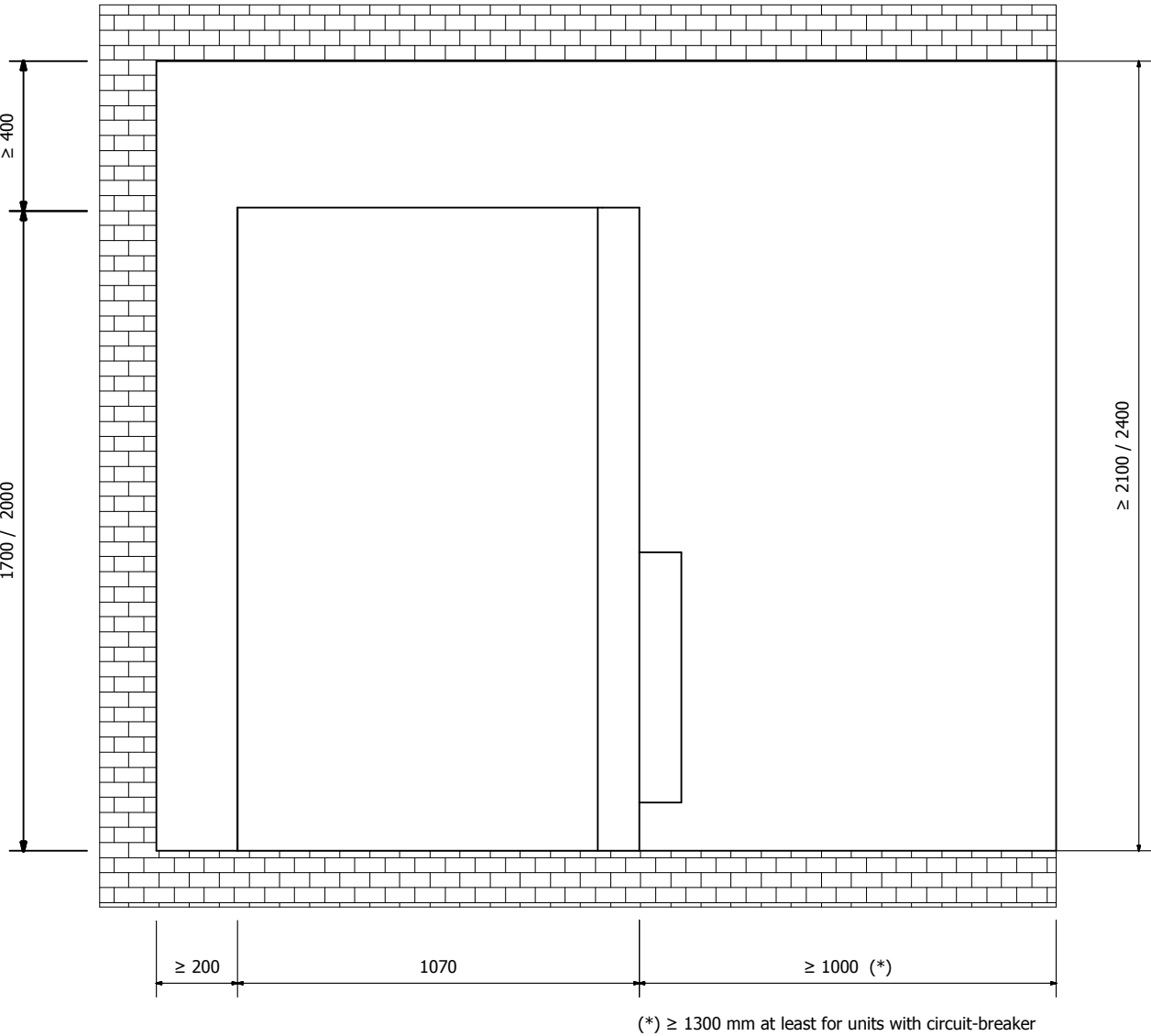
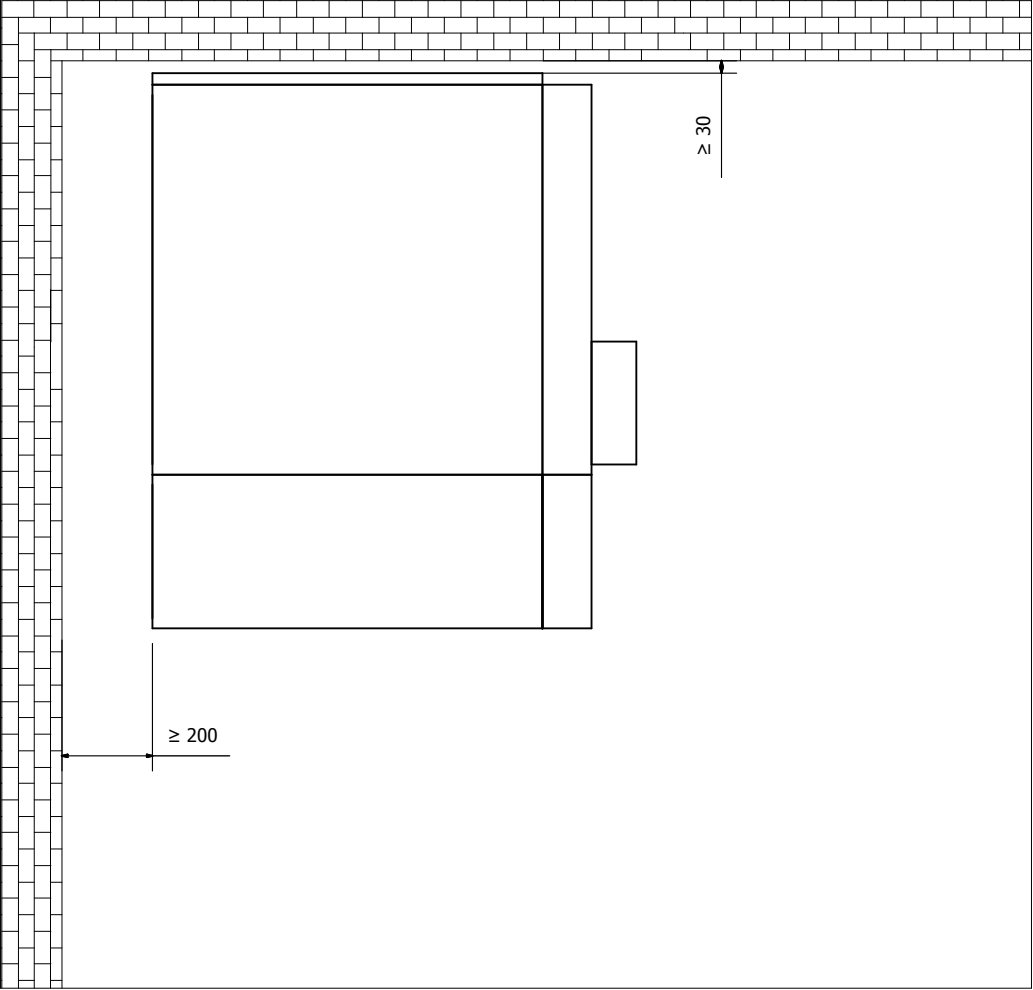
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RULES FOR INSTALLATION IN THE ROOM

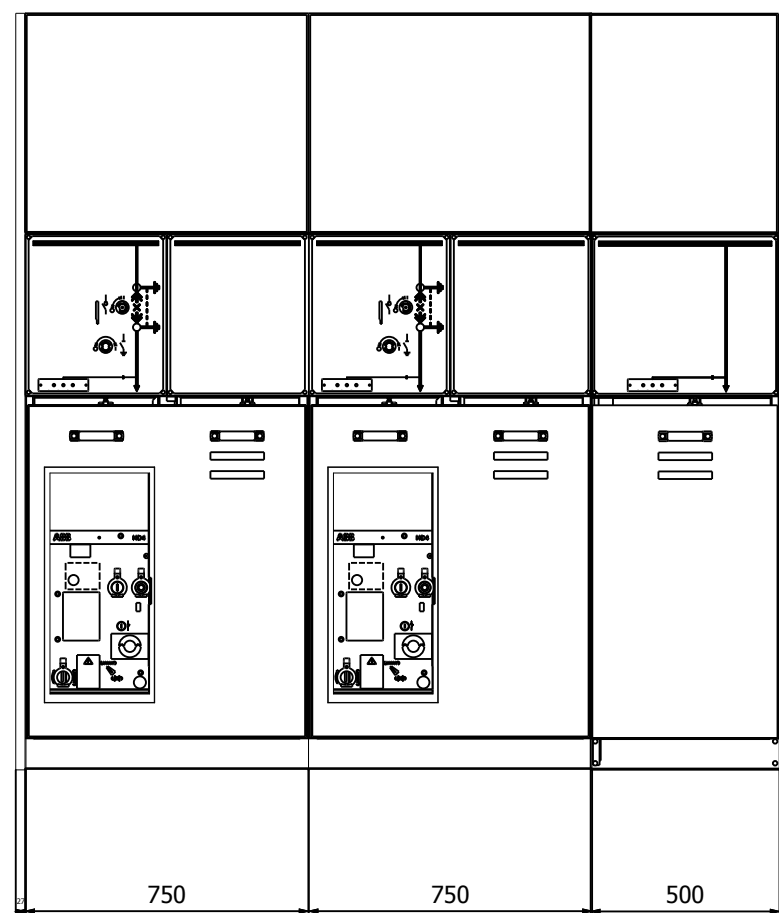
MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			INSTALLATION RULES SLD & GA Drawing MVD-28 20kV Central Office	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 9		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 16		
							PT. ABB Sakti Industri					

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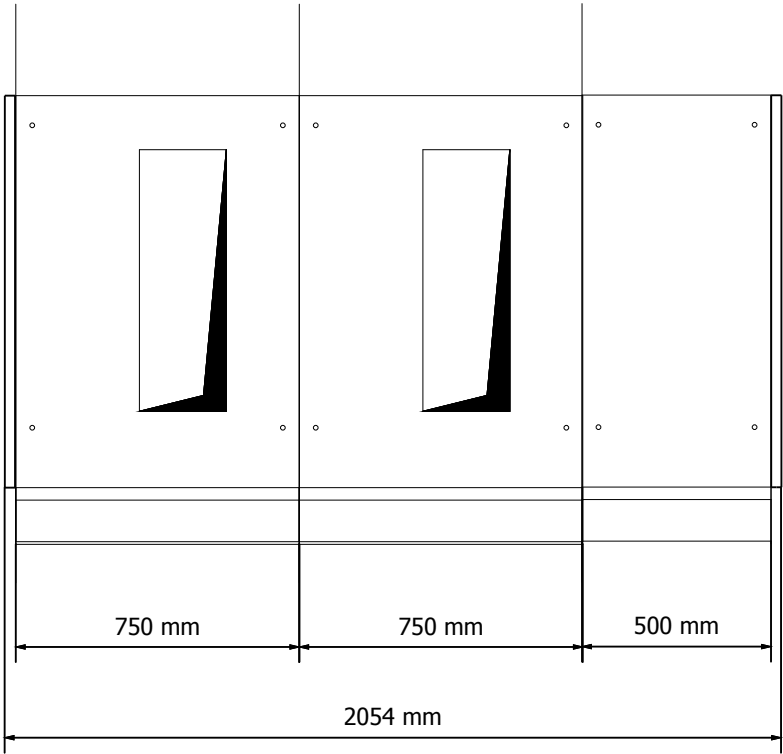


Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 630kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-28 20kV Central Office	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri		
0	First Issued	19.11.2018	YR	Checked	AW							
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042						
											Customer	PT Multimas Nabati Asahan
									Drawing Nbr.		Sh.	10
									Customer Doc. Nbr.		Cont.	16

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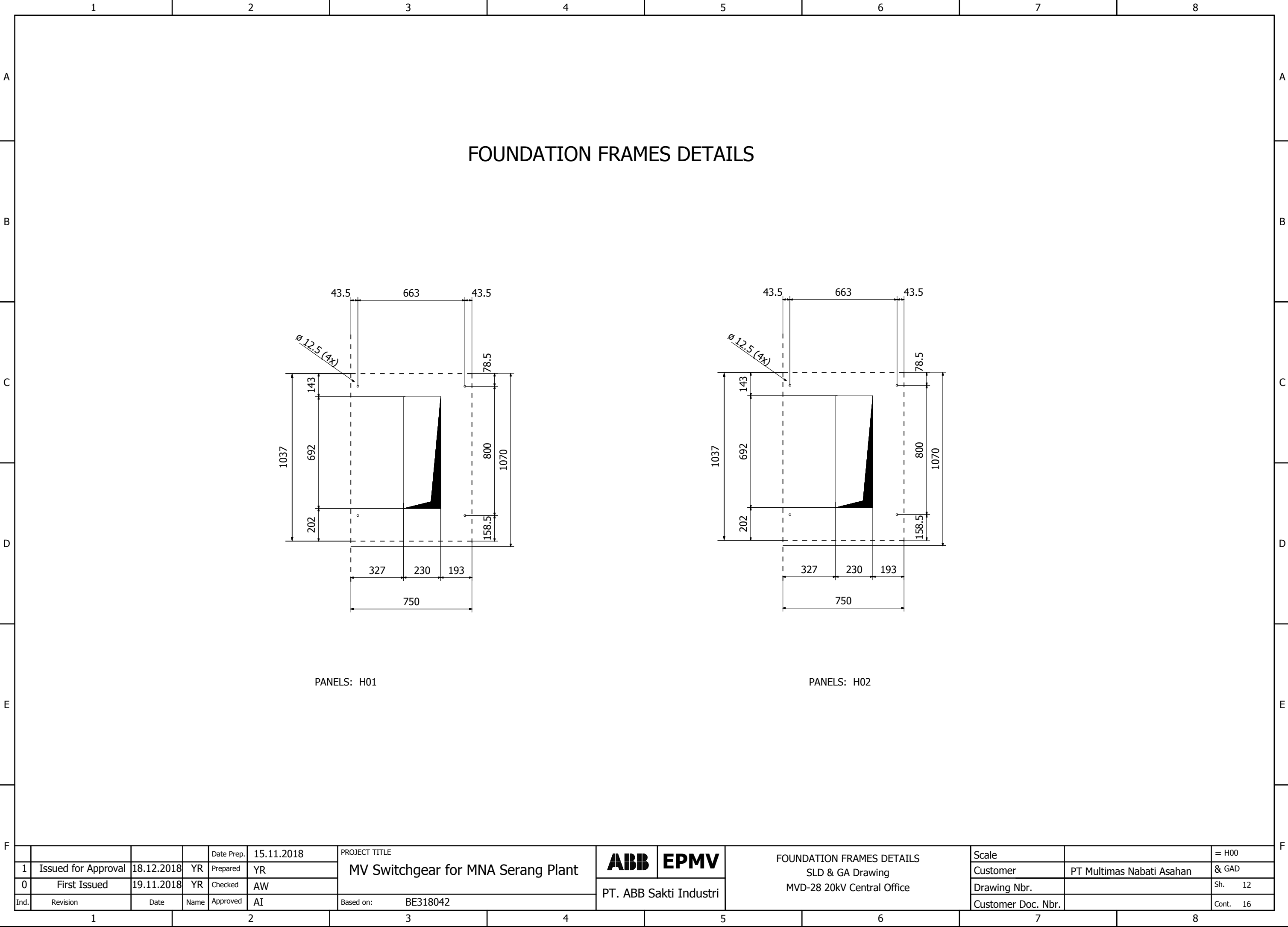
FOUNDATION FRAMES



Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 630kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-28 20kV Central Office	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 11		
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-28 20kV Central Office		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR						Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW						Drawing Nbr.		Sh. 12
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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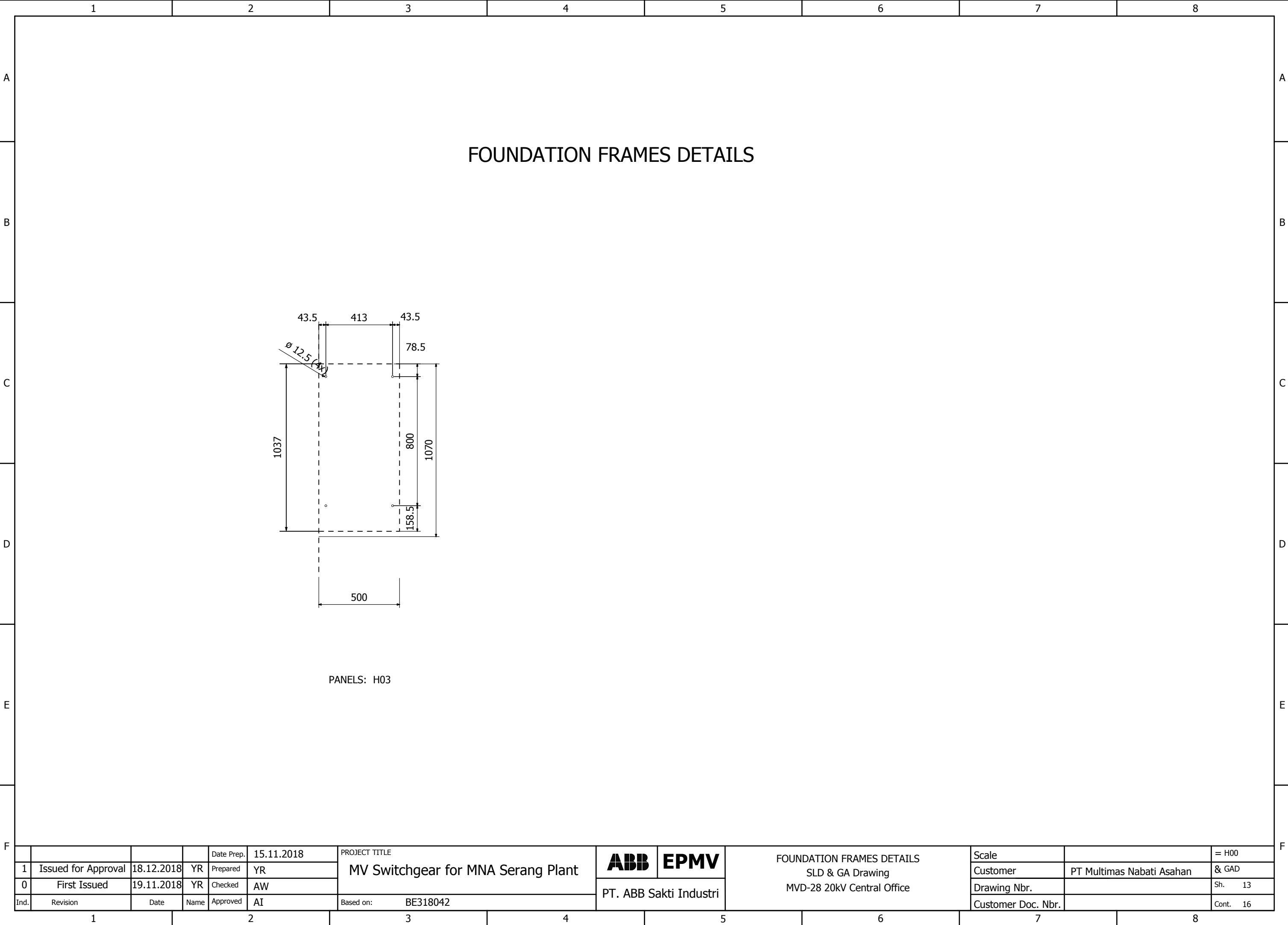
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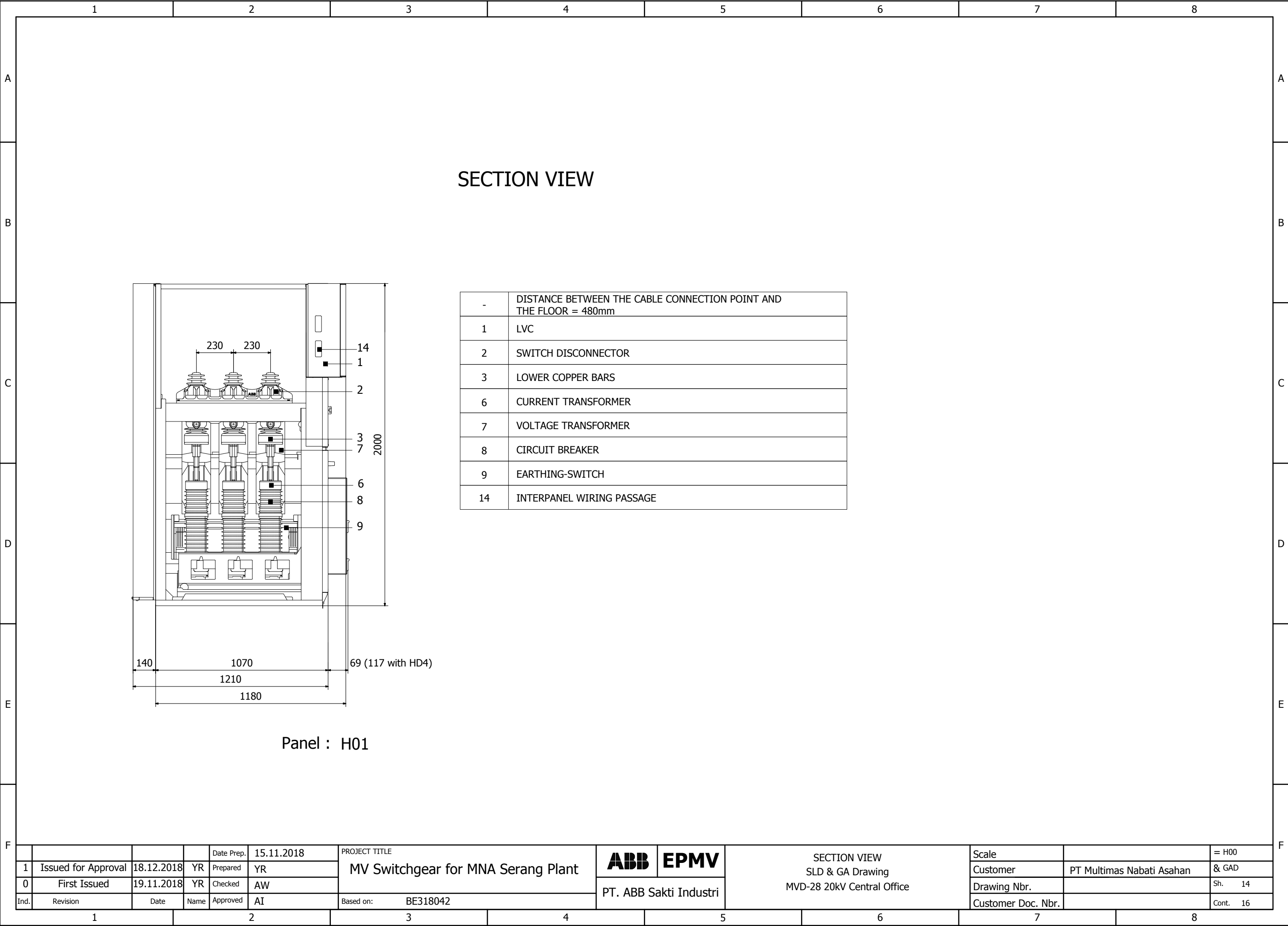
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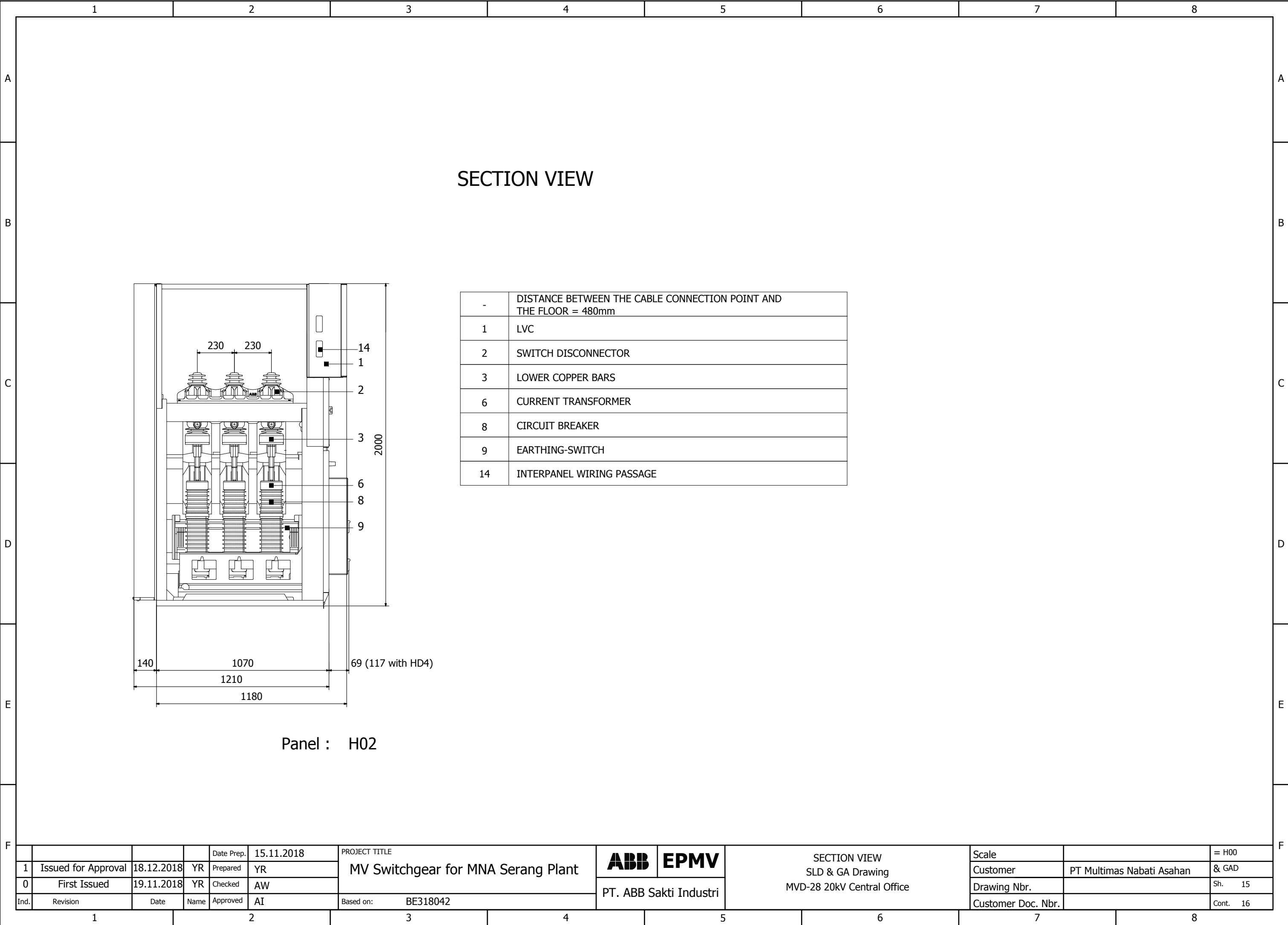


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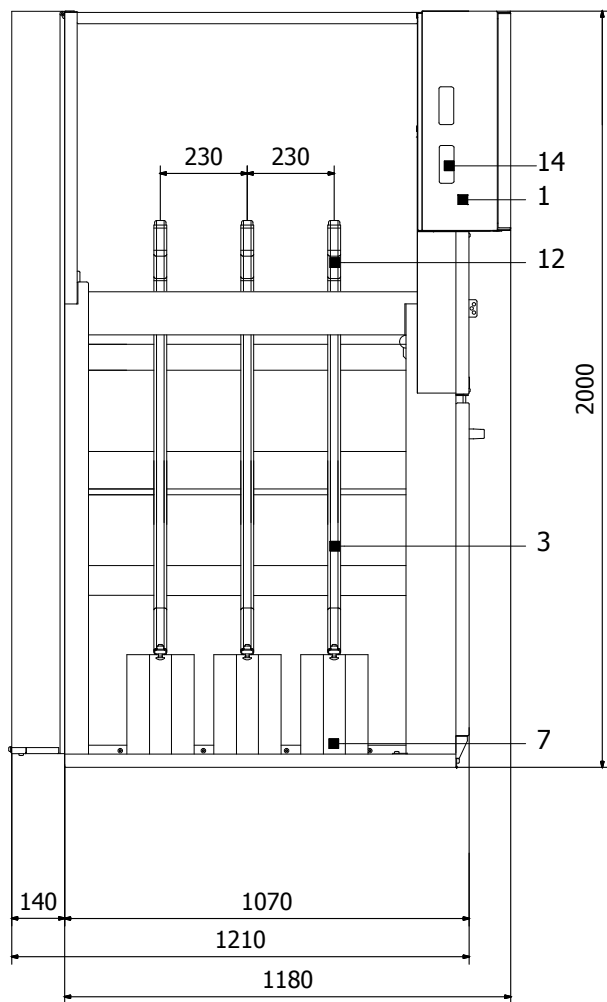
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-28 20kV Central Office	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 14		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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SECTION VIEW



-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 670mm
1	LVC
3	LOWER COPPER BARS
6	VOLTAGE TRANSFORMER
12	TOP COPPER BARS
14	INTERPANEL WIRING PASSAGE

Panel : H03

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	SECTION VIEW SLD & GA Drawing MVD-28 20kV Central Office		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR						Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW	Based on: BE318042		PT. ABB Sakti Industri			Drawing Nbr.		Sh. 16
Ind.	Revision	Date	Name	Approved	AI						Customer Doc. Nbr.		Cont. 16